

① Consider the non-linear first order ODEs and

(i) Solve $\frac{dx}{dt} = x - x^2$ with $[x(0) = 2]$

(ii) Find the steady states and their stability

(iii) What happens to the solution from the initial state $x(0) = 2$ as $t \rightarrow \infty$? Explain graphically showing the steady states.

Solution

Note: The general form of the non-linear, first-order ODE is $\frac{dx}{dt} = ax - bx^2$.

(i) **Solution by letting $z = \frac{1}{x}$**

Let $z = \frac{1}{x} \therefore \frac{dz}{dt} = -\frac{1}{x^2} \frac{dx}{dt}$

Recall ODE: $\frac{dx}{dt} = x - x^2$

Substitute $\frac{dx}{dt}$ in $\frac{dz}{dt} = -\frac{1}{x^2} \frac{dx}{dt}$

$$\Rightarrow \frac{dz}{dt} = \left(-\frac{1}{x^2}\right)(x - x^2)$$

$$= \left(-\frac{1}{x^2}\right)x + \left(\frac{1}{x^2}\right)x^2$$

$$\Rightarrow \frac{dz}{dt} = -\frac{1}{x} + 1 \quad \text{Recall: } z = \frac{1}{x}$$

$\therefore \frac{dz}{dt} = -z + 1$ The original non-linear, 1st order ODE is now a linear, 1st order ODE.

Particular solution (z_p):

let $z_p = A \quad \frac{dz_p}{dt} = -z_p + 1$

$$\Rightarrow \frac{dA}{dt} = -A + 1 \Rightarrow \boxed{A = 1}$$

$$\therefore \boxed{z_p = 1}$$

Homogenous solution (z_h):

$$\frac{dz_h}{dt} = -z_h$$

(i) **Separable equations based solution**

Given $\frac{dx}{dt} = x - x^2$

$$\Rightarrow \frac{dx}{x - x^2} = dt$$

Integrating

$$\Rightarrow \int \frac{dx}{x - x^2} = \int dt + C$$

Note: This integral needs to be addressed using partial fractions.

Consider the integrand $\frac{1}{x - x^2}$

$$\frac{1}{x - x^2} = \frac{1}{x(1 - x)}$$

In partial fractions:

$$\frac{1}{x(1 - x)} = \frac{A}{x} + \frac{B}{1 - x}$$

$$\frac{1}{x(1 - x)} = \frac{A(1 - x) + Bx}{x(1 - x)}$$

$$1 = A - x(A - B)$$

Comparing x^0 & x^1 terms

Solution with $z=1/x$ continued

$$\frac{dz_h}{z_h} = -1 dt$$

Integrating

$$\int \frac{dz_h}{z_h} = \int -1 dt + C$$

$$\ln(z_h) = -t + C$$

$$\Rightarrow z_h = e^{-t+C} \Rightarrow z_h = Ce^{-t}$$

Complete solution:

$$z = z_p + z_h$$

$$z(t) = 1 + Ce^{-t}$$

But we really need $x(t)$ to solve the ODE. Recall: $z = \frac{1}{x(t)}$

$$\therefore x(t) = \frac{1}{z(t)}$$

$$\Rightarrow x(t) = \frac{1}{1 + Ce^{-t}}$$

Apply initial condition: $x(0) = 2$

$$x(0) = 2 = \frac{1}{1 + Ce^0} = \frac{1}{1+C}$$

$$\Rightarrow 2 + 2C = 1 \Rightarrow C = -\frac{1}{2}$$

$$\therefore x(t) = \frac{1}{1 - \frac{1}{2}e^{-t}}$$

$$x(t) = \frac{2e^t}{2e^t - 1}$$

Separable equations-based solution continued

$$x: A = 1$$

$$x: -(A-B) = 0 \Rightarrow A = B$$

$$\Rightarrow B = 1$$

$$\therefore \frac{1}{x(1-x)} = \frac{1}{x} + \frac{1}{1-x}$$

$$\begin{aligned} \Rightarrow \int \frac{dx}{x(1-x)} &= \int \left(\frac{1}{x} + \frac{1}{1-x} \right) dx \\ &= \int \frac{1}{x} dx + \int \frac{1}{1-x} dx \\ &= \ln(x) - \ln(1-x) \end{aligned}$$

$$\int \frac{dx}{x(1-x)} = \ln\left(\frac{x}{1-x}\right)$$

Recall: $\int \frac{dx}{x(1-x)} = \int dt + C$

$$\Rightarrow \ln\left(\frac{x}{1-x}\right) = t + C$$

$$\Rightarrow \left(\frac{x}{1-x}\right) = e^{t+C} = Ce^t$$

$$\Rightarrow x = (1-x)Ce^t$$

$$\Rightarrow x = Ce^t - xCe^t$$

$$\Rightarrow x + xCe^t = Ce^t \Rightarrow x(1 + Ce^t) = Ce^t$$

$$\therefore x(t) = \frac{Ce^t}{1 + Ce^t}$$

Now Apply initial condition: $x(0) = 2$

$$x(0) = 2 = \frac{Ce^0}{Ce^0 + 1} \Rightarrow 2 = \frac{C}{C+1}$$

$$\Rightarrow 2C + 2 = C$$

$$\Rightarrow C = -2$$

$$\therefore x(t) = \frac{-2e^t}{-2e^t + 1} = \frac{-2e^t}{-(2e^t - 1)}$$

$$\Rightarrow x(t) = \frac{2e^t}{2e^t - 1}$$

(3)

KVB

(i) To find the steady states of $\frac{dx}{dt} = x - x^2$ and their stability:

Given $\frac{dx}{dt} = x - x^2$; Let $f = x - x^2$

Note:

Steady states occur where slope = 0 or $\frac{dx}{dt} = f = 0$

$$\Rightarrow x - x^2 = 0 \Rightarrow x(1-x) = 0 \therefore \boxed{x=0} \quad \boxed{x=1}$$

Two steady states

Note:

Stability of the steady states are assessed by $\frac{df}{dx}$

$$\textcircled{1} \left. \frac{df}{dx} \right|_{x=0} = \left. \frac{d}{dx}(x - x^2) \right|_{x=0} = (1 - 2x) \Big|_{x=0} = 1 - 2(0) = 1$$

$$\therefore \left. \frac{df}{dx} \right|_{x=0} = 1 > 0$$

$\therefore \left. \frac{df}{dx} \right|_{x=0} > 0$, $\boxed{x=0}$ is an unstable steady state

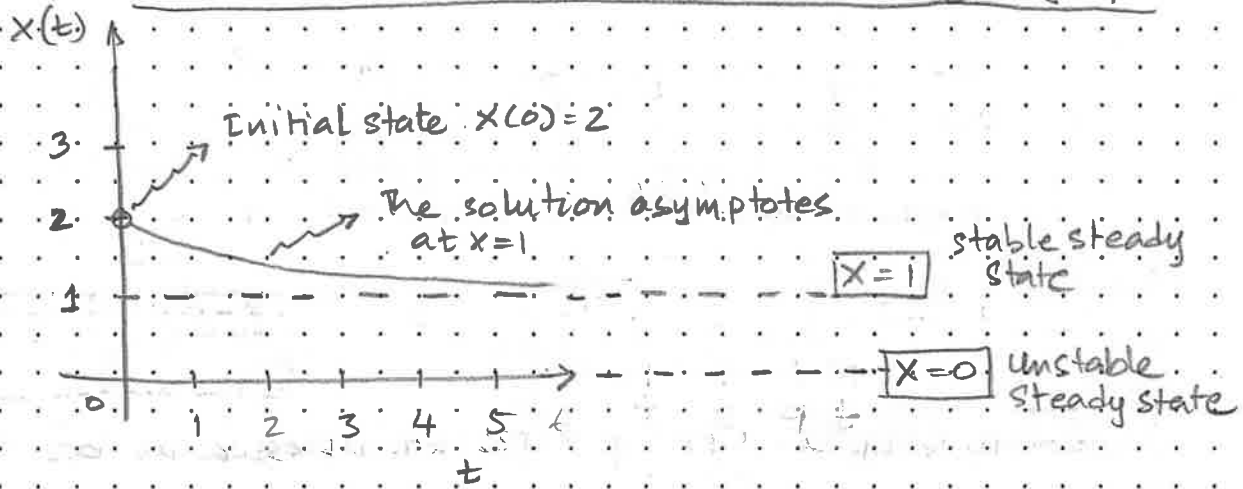
$$\textcircled{2} \left. \frac{df}{dx} \right|_{x=1}$$

$$= \left. \frac{d}{dx}(x - x^2) \right|_{x=1} = (1 - 2x) \Big|_{x=1} = 1 - 2(1) = -1$$

$$\therefore \left. \frac{df}{dx} \right|_{x=1} = -1 < 0$$

$\therefore \left. \frac{df}{dx} \right|_{x=1} < 0$, $\boxed{x=1}$ is a stable steady state

(iii) The development of the solution $x(t)$ from the initial state, $x(0) = 2$ using a graph:



② Consider the non-linear, first order ODE, and

(i) Solve $\frac{dx}{dt} = 3x(5-x)$ with $[x(0) = 8]$

(ii) Find the steady states and their stability

(iii) What happens to the solution from the initial state $x(0) = 8$ as $t \rightarrow \infty$? Explain graphically showing the steady states.

Solution Note: The general form of the non-linear, first order

ODE is $\frac{dx}{dt} = ax - bx^2$, $\therefore \frac{dx}{dt} = 15x - 3x^2$

(i)

Solution by letting $z = \frac{1}{x}$

Let $z = \frac{1}{x}$ $\therefore \frac{dz}{dt} = \left(-\frac{1}{x^2}\right) \left(\frac{dx}{dt}\right)$

Recall ODE: $\frac{dx}{dt} = 15x - 3x^2$

Substitute in $\frac{dz}{dt} = \left(-\frac{1}{x^2}\right) \left(\frac{dx}{dt}\right)$

$\Rightarrow \frac{dz}{dt} = \left(-\frac{1}{x^2}\right) (15x - 3x^2)$

$= -\frac{15x}{x^2} + \frac{3x^2}{x^2}$

$\Rightarrow \frac{dz}{dt} = -\frac{15}{x} + 3$ Recall: $z = \frac{1}{x}$

$\therefore \frac{dz}{dt} = -15z + 3$ The original non-linear, 1st order ODE is now a linear 1st order ODE

Particular solution (z_p):

Let $z_p = A$ $\frac{dz_p}{dt} = -15z_p + 3$

$\Rightarrow \frac{dA}{dt} = -15A + 3 \Rightarrow 15A = 3$

$\Rightarrow A = \frac{3}{15} \Rightarrow A = \frac{1}{5}$

$\therefore z_p = \frac{1}{5}$

(ii)

Separable equation-based solution

Given $\frac{dx}{dt} = 15x - 3x^2$

$\Rightarrow \frac{dx}{dt} = 3x(5-x)$ Original form of ODE

$\Rightarrow \frac{dx}{3x(5-x)} = dt$

Integrating

$\Rightarrow \int \frac{dx}{3x(5-x)} = \int dt + C$

Note: This integral needs to be addressed using partial fractions

Consider the integrand, $\frac{1}{3x(5-x)}$

In partial fractions:

$\frac{1}{3x(5-x)} = \frac{A}{3x} + \frac{B}{5-x}$

$\frac{1}{3x(5-x)} = \frac{5A - Ax + 3Bx}{3x(5-x)}$

$1 = 5A - (A - 3B)x$

Comparing x^0 & x^1 terms

Solution with $z = \frac{1}{x}$ continued

Homogenous solution (z_h):

$$\frac{dz_h}{dt} = -15z_h$$

$$\frac{dz_h}{z_h} = -15 dt$$

Integrating

$$\int \frac{dz_h}{z_h} = \int -15 dt + C$$

$$\ln(z_h) = -15t + C$$

$$z_h = e^{-15t+C} = C e^{-15t}$$

$$\Rightarrow z_h = C e^{-15t}$$

Complete solution: $z = z_p + z_h$

$$z(t) = \frac{1}{5} + C e^{-15t}$$

But we really need $x(t)$ to solve the ODE. Recall $z(t) = \frac{1}{x(t)}$

$$x(t) = \frac{1}{\frac{1}{5} + C e^{-15t}}$$

$$x(t) = \frac{5}{1 + 5C e^{-15t}}$$

Apply initial state or condition:

$$x(0) = \frac{5}{1 + 5C e^{-15(0)}} = 8 \quad [x(0) = 8]$$

$$= \frac{5}{1 + 5C} = 8 \Rightarrow 8 + 40C = 5 \Rightarrow C = \frac{-3}{40}$$

$$x(t) = \frac{5}{1 + 5(-3/40) e^{-15t}}$$

Separable equations-based solution continued

⑥

KVB

$$x^0: 5A = 1 \Rightarrow A = 1/5$$

$$x^1: -(A - 3B) = 0 \Rightarrow 3B = A$$

$$\Rightarrow B = \frac{A}{3} \Rightarrow B = \frac{1}{15}$$

$$\frac{1}{3x(5-x)} = \frac{1/5}{3x} + \frac{1/15}{5-x}$$

$$= \frac{1}{15x} + \frac{1}{15} \left(\frac{1}{5-x} \right)$$

$$\Rightarrow \int \frac{dx}{3x(5-x)} = \int \frac{1}{15x} dx + \int \frac{1}{15} \left(\frac{1}{5-x} \right) dx$$

$$= \frac{1}{15} \ln(x) + \frac{1}{15} \ln(5-x)$$

$$= \left(\frac{1}{15} \right) \ln \left(\frac{x}{5-x} \right)$$

Recall:

$$\int \frac{dx}{3x(5-x)} = \int dt + C$$

$$\Rightarrow \left(\frac{1}{15} \right) \ln \left(\frac{x}{5-x} \right) = t + C$$

$$\Rightarrow \ln \left(\frac{x}{5-x} \right) = 15t + 15C$$

$$\Rightarrow \left(\frac{x}{5-x} \right) = e^{15t + 15C} = C e^{15t}$$

$$\Rightarrow x = (5-x) C e^{15t} = 5C e^{15t} - x C e^{15t}$$

$$\Rightarrow x + x C e^{15t} = 5C e^{15t}$$

$$\Rightarrow x(1 + C e^{15t}) = 5C e^{15t}$$

$$\Rightarrow x(t) = \frac{5C e^{15t}}{1 + C e^{15t}}$$

Apply initial state or condition:

$$x(0) = \frac{5C e^{15(0)}}{1 + C e^{15(0)}} = 8 \quad [x(0) = 8]$$

Solution with $z = \frac{1}{x}$ continued.

$$x(t) = \frac{5}{1 - \left(\frac{3}{8}\right)e^{-15t}}$$

Separable equation - based.
Solution continued.

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KVB

$$\Rightarrow \frac{5C}{1+C} = 8$$

$$\Rightarrow 8 + 8C = 5C \Rightarrow C = -\frac{8}{3}$$

$$\begin{aligned} x(t) &= \frac{5 \left(-\frac{8}{3}\right) e^{15t}}{1 + \left(-\frac{8}{3}\right) e^{15t}} \\ &= \frac{\left(-\frac{8}{3}\right) \cdot 5 \cdot e^{15t}}{\left(-\frac{8}{3}\right) \left[-\frac{3}{8} + e^{15t}\right]} \\ &= \frac{5}{e^{-15t} \left(e^{15t} - \frac{3}{8}\right)} \end{aligned}$$

$$x(t) = \frac{5}{1 - \left(\frac{3}{8}\right)e^{-15t}}$$

(ii) To find the steady states of $\frac{dx}{dt} = 3x(5-x)$ and their stability:

$$\text{Given } \frac{dx}{dt} = 3x(5-x) \Rightarrow \frac{dx}{dt} = x(15-3x)$$

$$\text{Let } f = x(15-3x)$$

Note: Steady states occur where slope = 0 or $\frac{dx}{dt} = f = 0$

$$\Rightarrow x(15-3x) = 0 \Rightarrow \boxed{x=0} \text{ \& } (15-3x)=0 \Rightarrow \boxed{x=5}$$

Two steady states

Note: Stability of steady states are assessed by $\frac{df}{dx}$

$$\begin{aligned} \textcircled{1} \quad \left. \frac{df}{dx} \right|_{x=0} &= \left. \frac{d}{dx} [x(15-3x)] \right|_{x=0} = \left. \frac{d}{dx} (15x - 3x^2) \right|_{x=0} \\ &= (15 - 6x)|_{x=0} \\ &= 15 - 0 = 15 \end{aligned}$$

$$\therefore \left. \frac{df}{dx} \right|_{x=0} = 15 > 0$$

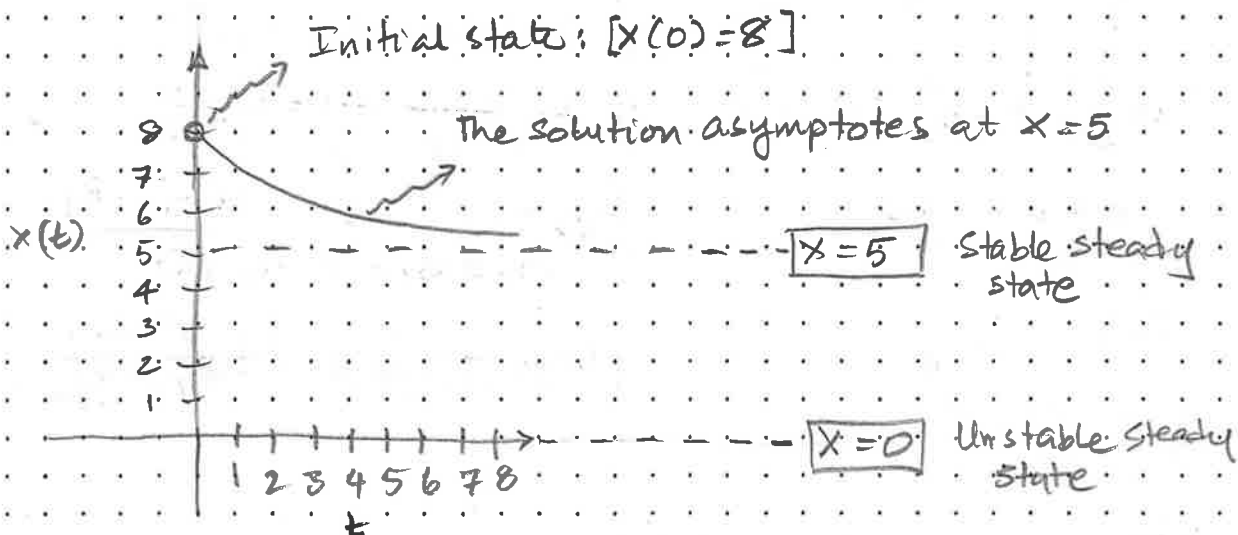
$\therefore \left. \frac{df}{dx} \right|_{x=0} > 0$ $x=0$ is an unstable steady state

$$\begin{aligned} \textcircled{2} \quad \left. \frac{df}{dx} \right|_{x=5} &= \left. \frac{d}{dx} [x(15-3x)] \right|_{x=5} = \left. \frac{d}{dx} (15x - 3x^2) \right|_{x=5} \\ &= (15 - 6x)|_{x=5} \\ &= 15 - 30 = -15 \end{aligned}$$

$$\therefore \left. \frac{df}{dx} \right|_{x=5} = -15 < 0$$

$\therefore \left. \frac{df}{dx} \right|_{x=5} < 0$ $x=5$ is a stable steady state

(iii) The development of the solution $x(t)$ from the initial state, $x(0) = 8$ using a graph:



③ Consider the following non-linear, first order ODE

$$\frac{dy}{dt} = y - y^3$$

Solve it and discuss its steady states, and stability for the following initial states: $[y(0) = 3, -3, 0.5, -0.5]$

Solution by letting $z = \frac{1}{y}$

$$\text{Let } z = \frac{1}{y} \quad \therefore \frac{dz}{dt} = -\frac{1}{y^2} \frac{dy}{dt}$$

Recall ODE: $\frac{dy}{dt} = y - y^3$

Substitute in: $\frac{dz}{dt} = -\frac{1}{y^2} \frac{dy}{dt}$

$$\Rightarrow \frac{dz}{dt} = \left(-\frac{1}{y^2}\right)(y - y^3)$$

$$= -\frac{y}{y^2} + \frac{y^3}{y^2}$$

$$\Rightarrow \frac{dz}{dt} = -\frac{1}{y} + y \quad \text{Recall: } z = \frac{1}{y}$$

$$\therefore \frac{dz}{dt} = -z + \frac{1}{z} = \frac{-z^2 + 1}{z}$$

$$\Rightarrow \frac{dz}{dt} = \frac{1 - z^2}{z}$$

$$\Rightarrow \frac{z dz}{1 - z^2} = dt$$

Integrating

$$\int \frac{z dz}{1 - z^2} = \int dt + C$$

Note: This integral can be addressed by method of substitution

Separable equation-based solution

Given ODE: $\frac{dy}{dt} = y - y^3$

$$\Rightarrow \frac{dy}{y - y^3} = dt$$

$$\Rightarrow \frac{dy}{y(1 - y^2)} = dt$$

$$\Rightarrow \frac{dy}{(y)(1 - y)(1 + y)} = dt$$

Integrating

$$\Rightarrow \int \frac{dy}{(y)(1 - y)(1 + y)} = \int dt + C$$

Note: This integral needs to be addressed using Partial Fractions

Consider the integrand:

$$\frac{1}{(y)(1 - y)(1 + y)} = \frac{A}{y} + \frac{B}{1 - y} + \frac{C}{1 + y}$$

$$\Rightarrow \frac{1}{(y)(1 - y)(1 + y)} = \frac{A(1 - y^2) + By(1 + y) + Cy(1 - y)}{(y)(1 - y)(1 + y)}$$

$$\Rightarrow \frac{1}{(y)(1 - y)(1 + y)} = \frac{A - Ay^2 + By + By^2 + Cy - Cy^2}{(y)(1 - y)(1 + y)}$$

$$\Rightarrow 1 = A + (B + C)y - (A - B + C)y^2$$

Comparing terms, y^0, y^1, y^2 we have

Solution by letting $z = \frac{1}{y}$
Continued

Consider the integral, $\int \frac{z dz}{1-z^2}$

$$\text{Let } 1-z^2 = u \Rightarrow du = -2z dz$$

$$\Rightarrow z dz = -\frac{du}{2}$$

Substitute u , & $\frac{du}{2}$ in $\int \frac{z dz}{1-z^2}$

$$\int \frac{z dz}{1-z^2} = \int \frac{-\frac{du}{2}}{2u} = -\frac{1}{2} \int \frac{du}{u}$$

$$= -\frac{1}{2} \ln(u)$$

$$= -\frac{1}{2} \ln(1-z^2)$$

Recall: $\int \frac{z dz}{1-z^2} = \int dt + C$

$$\Rightarrow -\frac{1}{2} \ln(1-z^2) = t + C$$

$$\Rightarrow \ln(1-z^2)^{-1/2} = t + C$$

$$\Rightarrow \frac{1}{\sqrt{1-z^2}} = e^{t+C} = e^t \cdot e^C$$

$$\Rightarrow \frac{1}{\sqrt{1-z^2}} = C e^t$$

$$\Rightarrow \frac{1}{1-z^2} = C e^{2t}$$

$$\Rightarrow C e^{2t} - z^2 C e^{2t} = 1$$

$$\Rightarrow z^2 = \frac{C e^{2t} - 1}{C e^{2t}}$$

$$\Rightarrow \boxed{z(t) = \pm \left(\frac{C e^{2t} - 1}{C e^{2t}} \right)^{1/2}}$$

But we really need $y(t)$ to solve the ODE. Recall $z(t) = \frac{1}{y(t)}$

Separable equation-based solution continued

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KVB

$$y^0: \boxed{A=1}$$

$$y^1: B+C=0$$

$$y^2: -(A-B+C)=0$$

$$\Rightarrow 1=B-C$$

$$\Rightarrow \boxed{B=\frac{1}{2}}$$

$$\Rightarrow \boxed{C=-\frac{1}{2}}$$

$$\begin{aligned} B-C &= 1 \\ B+C &= 0 \end{aligned}$$

$$2B=1$$

$$\therefore \frac{1}{(y)(1-y)(1+y)} = \frac{1}{y} + \frac{1}{2(1-y)} - \frac{1}{2(1+y)}$$

$$\Rightarrow \frac{1}{y-y^3} = \frac{1}{y} + \frac{1}{2(1-y)} - \frac{1}{2(1+y)}$$

$$\begin{aligned} \Rightarrow \int \frac{dy}{y-y^3} &= \int \frac{dy}{y} + \int \frac{dy}{2(1-y)} - \int \frac{dy}{2(1+y)} \\ &= \ln(y) - \ln(1-y)^{1/2} - \ln(1+y)^{1/2} \\ &= \ln\left(\frac{y}{\sqrt{1-y^2}}\right) \end{aligned}$$

Recall: $\int \frac{dy}{y-y^3} = \int dt + C$

$$\Rightarrow \ln\left(\frac{y}{\sqrt{1-y^2}}\right) = t + C$$

$$\Rightarrow \frac{y}{\sqrt{1-y^2}} = e^{t+C} = C e^t$$

$$\Rightarrow \frac{y^2}{1-y^2} = C e^{2t}$$

$$\Rightarrow y^2 = C e^{2t} - y^2 C e^{2t}$$

$$\Rightarrow y^2(1 + C e^{2t}) = C e^{2t}$$

$$\Rightarrow y^2 = \frac{C e^{2t}}{1 + C e^{2t}}$$

$$\Rightarrow y(t) = \pm \left(\frac{C e^{2t}}{1 + C e^{2t}} \right)^{1/2}$$

Solution by letting $z = \frac{1}{y}$
Continued

$$y(t) = \frac{1}{z(t)} = \pm \left(\frac{Ce^{2t}}{Ce^{2t} + 1} \right)^{1/2}$$

$$\Rightarrow y(t) = \pm \left(\frac{1}{1 + Ce^{-2t}} \right)^{1/2}$$

$$\Rightarrow y(t) = \pm \frac{1}{\sqrt{1 + Ce^{-2t}}}$$

separable equation-based
solution

$$y(t) = \pm \left(\frac{1}{Ce^{-2t} + 1} \right)^{1/2}$$

$$\Rightarrow y(t) = \pm \frac{1}{\sqrt{1 + Ce^{-2t}}}$$

(i) To find the steady states of $\frac{dy}{dt} = y - y^3$ and their stability:

Given ODE; $\frac{dy}{dt} = y - y^3$ Let $f = y - y^3$

Note: Steady states occur where slope = 0 or $\frac{dy}{dt} = f = 0$

$$\Rightarrow y - y^3 = 0 \Rightarrow y(1 - y)(1 + y) = 0 \Rightarrow \boxed{y=0} \quad \boxed{y=1} \quad \boxed{y=-1}$$

Three steady states

Note: Stability of steady states are assessed by $\frac{df}{dy}$

$$\textcircled{1} \left. \frac{df}{dy} \right|_{y=0} = \left. \frac{d}{dy} (y - y^3) \right|_{y=0} = (1 - 3y^2) \Big|_{y=0} = 1 - 0 = 1$$

$$\therefore \left. \frac{df}{dy} \right|_{y=0} = 1 > 0$$

$\therefore \left. \frac{df}{dy} \right|_{y=0} > 0$ $\boxed{y=0}$ is an unstable steady state

$$\textcircled{2} \quad \left. \frac{df}{dy} \right|_{y=1} = \left. \frac{d}{dy}(y-y^3) \right|_{y=1} = (1-3y^2) \Big|_{y=1} = 1-3 = -2$$

$$\therefore \left. \frac{df}{dy} \right|_{y=1} = -2 < 0$$

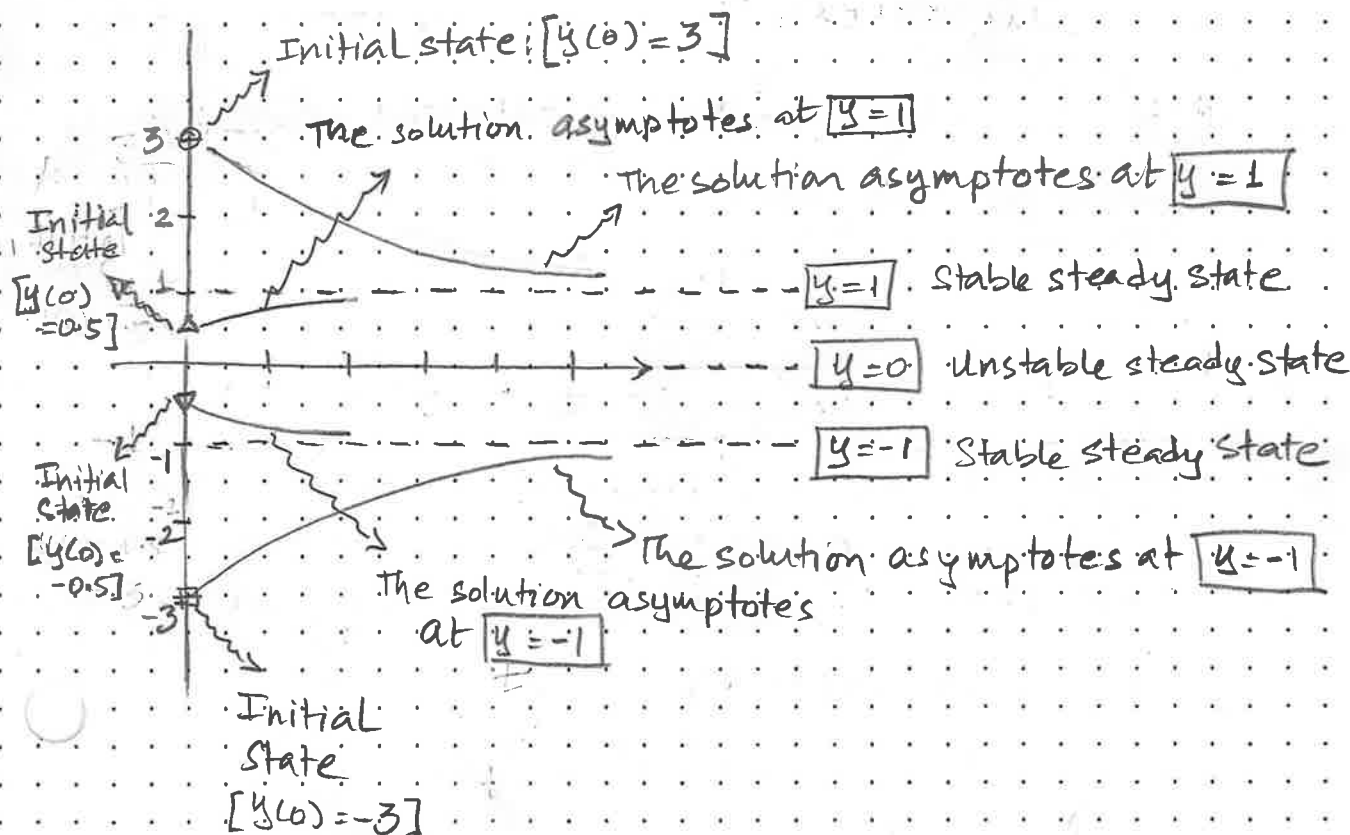
$\therefore \left. \frac{df}{dy} \right|_{y=1} < 0$ $y=1$ is a stable steady state

$$\textcircled{3} \quad \left. \frac{df}{dy} \right|_{y=-1} = \left. \frac{d}{dy}(y-y^3) \right|_{y=-1} = (1-3y^2) \Big|_{y=-1} = 1-3(-1)^2 = -2$$

$$\therefore \left. \frac{df}{dy} \right|_{y=-1} = -2 < 0$$

$\therefore \left. \frac{df}{dy} \right|_{y=-1} < 0$ $y=-1$ is a stable steady state

(iii) The development of the solution $y(t)$ from the initial states, $y(0) = 3, -3, 0.5, -0.5$:



- (13) KVB
 ④ Solve the following non-linear, first order ODE and discuss its steady states, and stability with the initial state $[x(0) = 3]$.

$$\boxed{\frac{dx}{dt} = 1 - x^2}$$

Solution: The given non-linear, first order ODE can easily be solved using the separable equations-based approach. Using the approach with $z = \frac{1}{x}$ gives it no advantage like with ODEs of the general form, $\left[\frac{dx}{dt} = ay - by^3\right]$.

Given ODE: $\frac{dx}{dt} = 1 - x^2 \Rightarrow \frac{dx}{1 - x^2} = dt$

Integrating

$$\Rightarrow \int \frac{dx}{1 - x^2} = \int dt + C$$

$$\Rightarrow \int \frac{dx}{(1-x)(1+x)} = \int dt + C$$

Note: This integral needs to be addressed using partial fractions.

Consider the integrand:

$$\frac{1}{(1-x)(1+x)} = \frac{A}{1-x} + \frac{B}{1+x}$$

$$\Rightarrow \frac{1}{(1-x)(1+x)} = \frac{A(1+x) + B(1-x)}{(1-x)(1+x)}$$

$$\Rightarrow \frac{1}{(1-x)(1+x)} = \frac{A + Ax + B - Bx}{(1-x)(1+x)}$$

$$\Rightarrow 1 = (A+B) + x(A-B)$$

Comparing the terms, x^0 , x^1 we have

$$x^0: A+B = 1$$

$$x^1: A-B = 0$$

$$2A = 1 \Rightarrow \boxed{A = \frac{1}{2}} \Rightarrow \boxed{B = \frac{1}{2}}$$

$$\frac{1}{(1-x)(1+x)} = \frac{1}{2(1-x)} + \frac{1}{2(1+x)}$$

$$\frac{dx}{(1-x)(1+x)} = \frac{dx}{2(1-x)} + \frac{dx}{2(1+x)}$$

$$\Rightarrow \frac{dx}{1-x^2} = \left(\frac{1}{2}\right) \frac{dx}{1-x} + \left(\frac{1}{2}\right) \frac{dx}{1+x}$$

Integrating

$$\Rightarrow \int \frac{dx}{1-x^2} = \left(\frac{1}{2}\right) \int \frac{dx}{1-x} + \left(\frac{1}{2}\right) \int \frac{dx}{1+x}$$

$$= \left(-\frac{1}{2}\right) \ln(1-x) + \left(\frac{1}{2}\right) \ln(1+x) = \left(\frac{1}{2}\right) \ln\left(\frac{1+x}{1-x}\right)$$

Recall: $\int \frac{dx}{1-x^2} = \int dt + C$

$$\Rightarrow \left(\frac{1}{2}\right) \ln\left(\frac{1+x}{1-x}\right) = t + C$$

$$\Rightarrow \ln\left(\frac{1+x}{1-x}\right) = 2t + C \Rightarrow \left(\frac{1+x}{1-x}\right) = e^{2t+C}$$

$$\Rightarrow \left(\frac{1+x}{1-x}\right) = Ce^{2t} \Rightarrow (1+x) = (1-x)Ce^{2t}$$

$$\Rightarrow (1+x) = Ce^{2t} - xCe^{2t}$$

$$\Rightarrow x + xCe^{2t} = Ce^{2t} - 1$$

$$\Rightarrow (Ce^{2t} + 1)x = (Ce^{2t} - 1)$$

$$\Rightarrow \boxed{x(t) = \frac{Ce^{2t} - 1}{Ce^{2t} + 1}}$$

Apply initial state: $[x(0) = 3]$

$$x(0) = 3 = \frac{Ce^{2 \cdot 0} - 1}{Ce^{2 \cdot 0} + 1} \Rightarrow 3 = \frac{C-1}{C+1} \Rightarrow 3C+3 = C-1 \Rightarrow 2C = -4 \Rightarrow \boxed{C = -2}$$

$$x(t) = \frac{-2e^{2t} - 1}{-2e^{2t} + 1}$$

$$\Rightarrow x(t) = \frac{-(2e^{2t} + 1)}{-(2e^{2t} - 1)}$$

$$\Rightarrow \boxed{x(t) = \frac{2e^{2t} + 1}{2e^{2t} - 1}}$$

To find the steady states of $\frac{dx}{dt} = 1 - x^2$ and their stability:

Given ODE: $\boxed{\frac{dx}{dt} = 1 - x^2}$ Let $f = 1 - x^2$

Note: Steady states occur where slope = 0 or

$$\frac{dx}{dt} = f = 0$$

$$\Rightarrow 1 - x^2 = 0 \Rightarrow (1 - x)(1 + x) = 0 \Rightarrow \boxed{x = 1} \quad \boxed{x = -1}$$

Two steady states

Note: Stability of steady states is assessed by $\frac{df}{dx}$

$$\textcircled{1} \left. \frac{df}{dx} \right|_{x=1} = \left. \frac{d}{dx}(1 - x^2) \right|_{x=1} = -2x \Big|_{x=1} = -2(1) = -2$$

$$\left. \frac{df}{dx} \right|_{x=1} = -2 < 0$$

$\therefore \left. \frac{df}{dx} \right|_{x=1} < 0$ $\boxed{x=1}$ is
a stable steady state.

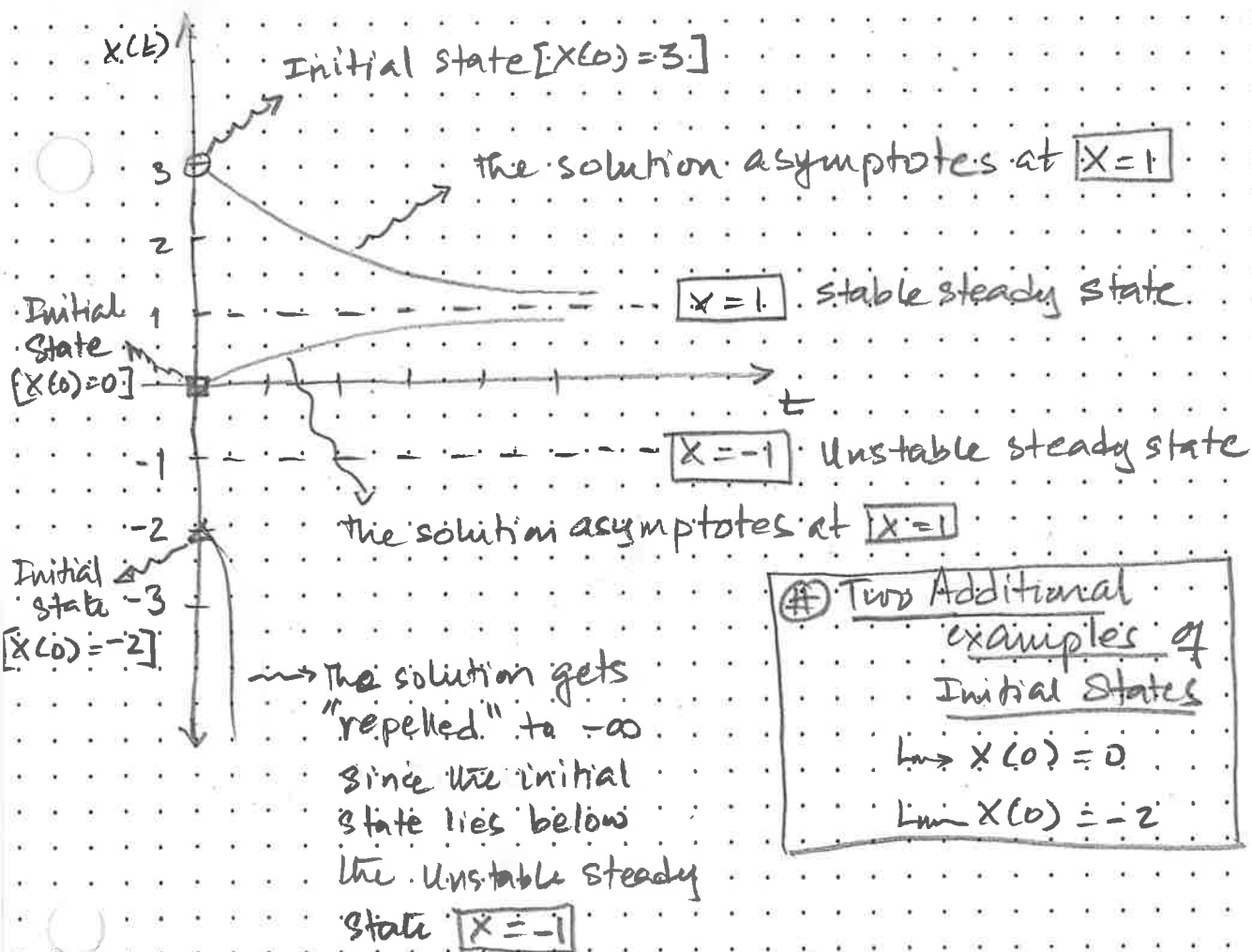
$$\textcircled{2} \left. \frac{df}{dx} \right|_{x=-1} = \left. \frac{df}{dx} (1-x^2) \right|_{x=-1}$$

$$= -2x \Big|_{x=-1} = (-2)(-1) = 2$$

$$\therefore \left. \frac{df}{dx} \right|_{x=-1} = 2 > 0$$

$\therefore \left. \frac{df}{dx} \right|_{x=-1} < 0$ $x = -1$ is
an unstable steady
state.

The development of the solution $x(t)$ from the
initial state, $[x(0) = 3]$:



Two Additional examples of Initial States

$$\lim_{t \rightarrow \infty} x(t) = 0$$

$$\lim_{t \rightarrow \infty} x(t) = -2$$